

Sin - Cos - Tan

Intro. To Trig

Notes and Assignment

Cah - Toa:
Trigonometry Notes
 etry?

properties of _____ and _____ or
 applications _____ you can find the missing _____ or
 angle by setting up trigonometric ratios
 ons as defined here work for _____

SOH - CAH - TOA

S - _____

O - _____

H - _____

C - _____

A - _____

H - _____

Introduction to Trigonometry Notes

Trigonometry?

study of the properties of triangles and trigonometric applications

, Cosine or Tangent you can find the missing side of a triangle by setting up trigonometric ratios
 g functions as defined here work for RIGHT

ine
 re of the leg opposite of
 use in a right triangle
 Sin = $\frac{\text{opposite}}{\text{hypotenuse}}$
 Sin A = $\frac{BC}{BA}$
 Sin B = $\frac{CA}{BA}$

leg adjacent to
 right triangle
 adjacent
 hypotenuse
 Cos = $\frac{\text{adjacent}}{\text{hypotenuse}}$
 Cos A = $\frac{CA}{BA}$
 Cos B = $\frac{BC}{BA}$

SOH - CAH - TOA

S - _____

O - _____

H - _____

C - _____

A - _____

H - _____

Introduction
 Part one: Find the ratio

$$\sin \angle A = \frac{8}{20} = \frac{2}{5}$$

$$\sin \angle P = \frac{30}{63} = \frac{10}{21}$$

$$\tan \angle A = \frac{8}{12} = \frac{2}{3}$$

$$\sin \angle B = \frac{12}{20} = \frac{3}{5}$$

Part Two: Find decimal place

$$\sin 51^\circ = 0.7761$$

$$\sin 26^\circ = 0.4384$$

$$\sin A = 0.9876$$

$$\tan S = 0.0349$$

$$\cos X = 0.4384$$

Secondary
 Math Shop

Soh - Cah - Toa:

Introduction to Trigonometry Notes

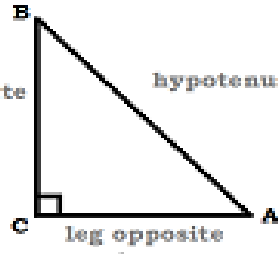
What is Trigonometry?

- It is the study of the properties of triangles and trigonometric functions and their applications
- Using Sine, Cosine or Tangent you can find the missing sides or angles of a right triangle by setting up trigonometric ratios
 - The Trig functions as defined here work for **RIGHT TRIANGLES**

ONLY

Sine

The ratio of the measure of the leg opposite of an angle to the hypotenuse in a right triangle



leg opposite $\angle A$

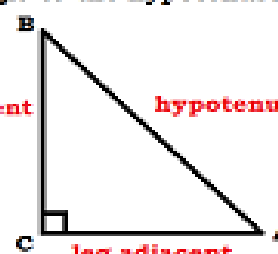
hypotenuse

leg opposite $\angle B$

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$
$$\sin A = \frac{BC}{BA}$$
$$\sin B = \frac{CA}{BA}$$

Cosine

The ratio of the measure of the leg adjacent to an angle to the hypotenuse in a right triangle



leg adjacent to $\angle B$

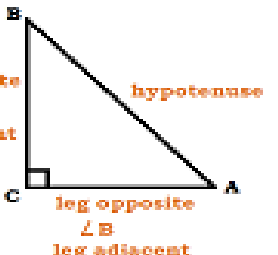
hypotenuse

leg adjacent to $\angle A$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$
$$\cos A = \frac{CA}{BA}$$
$$\cos B = \frac{BC}{BA}$$

Tangent

The ratio of the measurement of the leg opposite an angle to the measurement of the leg adjacent in a right triangle



leg opposite $\angle A$

leg adjacent to $\angle B$

hypotenuse

leg opposite $\angle B$

leg adjacent to $\angle A$

$$\tan = \frac{\text{opposite}}{\text{adjacent}}$$
$$\tan A = \frac{BC}{CA}$$
$$\tan B = \frac{CA}{BC}$$

SOH – CAH – TOA

S – Sine

O – Opposite

H – Hypotenuse

C – Cosine

A – Adjacent

H – Hypotenuse

T – Tangent

O – Opposite

A – Adjacent

Setting up Ratios:

- Each time you set up a Trig. Ratio it is set up in the form:

$$\text{trig ratio (angle)} = \frac{\text{side}}{\text{side}}$$

Examples: Find the ratio for each given Trig. Function.

$$\sin \angle M = \frac{16}{34} = \frac{8}{17}$$

$$\cos \angle Z = \frac{12}{20} = \frac{3}{5}$$

$$\tan \angle L = \frac{30}{16} = \frac{15}{8}$$

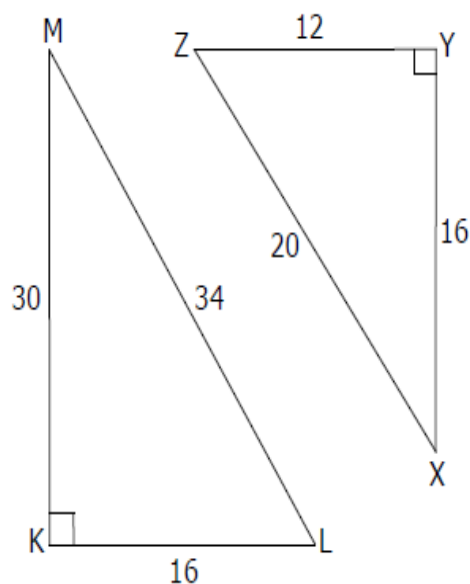
$$\tan \angle X = \frac{12}{16} = \frac{3}{4}$$

$$\cos \angle L = \frac{16}{34} = \frac{8}{17}$$

$$\tan \angle Z = \frac{16}{12} = \frac{4}{3}$$

$$\cos \angle M = \frac{30}{34} = \frac{15}{17}$$

$$\sin \angle X = \frac{12}{20} = \frac{3}{5}$$



Using the calculator:

- When you put a trig function into the calculator you want to make sure that your calculator is in degree mode
- If you simply want to find the value of the sin, cos or tan of an angle you can input it as $\sin(\text{angle})$, $\cos(\text{angle})$ or $\tan(\text{angle})$
- If you want to find out what angle has that trig value you will need to use the inverse trig function key on your calculator and put it in as $\sin^{-1}(\text{angle})$, $\cos^{-1}(\text{angle})$, or $\tan^{-1}(\text{angle})$

Examples: Find the value of each ratio. Round your answer to 4 places past the decimal place.

$$\sin 12^\circ = 0.2079 \quad \tan 74^\circ = 3.4874 \quad \sin 7^\circ = 0.1219$$

$$\cos 24^\circ = 0.9135 \quad \tan 54^\circ = 1.3764 \quad \sin 72^\circ = 0.9511$$

$$\cos 12^\circ = 0.9781 \quad \cos 20^\circ = 0.9397 \quad \tan 37^\circ = 0.7536$$

Examples: Find the measure of each angle to the nearest whole degree.

$$\sin A = 0.7245 = 46^\circ$$

$$\cos B = 0.2493 = 76^\circ$$

$$\tan C = 9.4618 = 84^\circ$$

$$\sin D = 0.4567 = 27^\circ$$

$$\cos E = 0.1212 = 83^\circ$$

$$\tan C = 0.4279 = 23^\circ$$

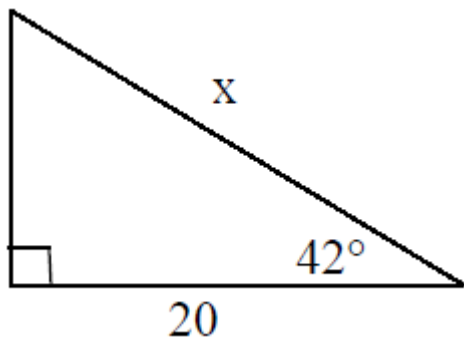
Solving Trig. Problems Using the Ratios:

- The majority of trigonometry problems that you encounter will fall into two categories:
 - Find the missing side given a side and an angle
 - Find the missing angle given two sides
 - Both of these can be solved by substituting into the equation:

$$\text{trig ratio (angle)} = \frac{\text{side}}{\text{side}}$$

- They will be solved in different ways however.

Finding the Missing Side Problems:



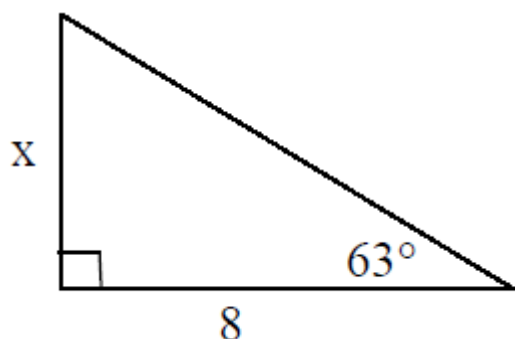
We are given the angle, the adjacent and the hypotenuse so that this is a cosine problem.

$$\cos 42 = \frac{20}{x} \quad \text{set-up trig ratio}$$

$$x (\cos 42) = 20 \quad \text{cross multiply}$$

$$x = \frac{20}{\cos 42} \quad \text{divide both sides by } \cos 42$$

$$x = 26.91 \quad \text{put in calculator}$$

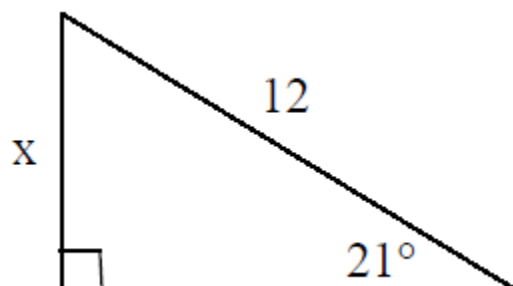


We are given the angle, the adjacent and the opposite so that this is a tangent problem.

$$\tan 63 = \frac{x}{8} \quad \text{set-up trig ratio}$$

$$8 (\tan 63) = x \quad \text{cross multiply}$$

$$15.7 = x \quad \text{put in calculator}$$



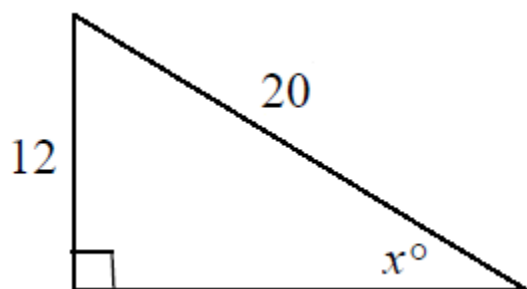
We are given the angle, the opposite and the hypotenuse so that this is a sine problem.

$$\sin 21 = \frac{x}{12} \quad \text{set-up trig ratio}$$

$$12 (\sin 21) = x \quad \text{cross multiply}$$

$$4.3 = x \quad \text{put in calculator}$$

Finding the Missing Angle Problems:

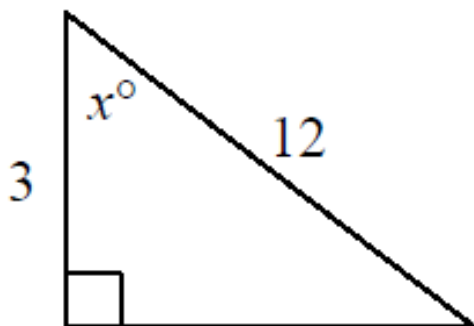


We are given the angle, the opposite and the hypotenuse so that this is a sine problem.

$$\sin x = \frac{12}{20} \quad \text{set-up trig ratio}$$

$$x = \sin^{-1}\left(\frac{12}{20}\right) \quad \text{use the inverse trig function to find angle}$$

$$x = 37^\circ \quad \text{put in calculator}$$

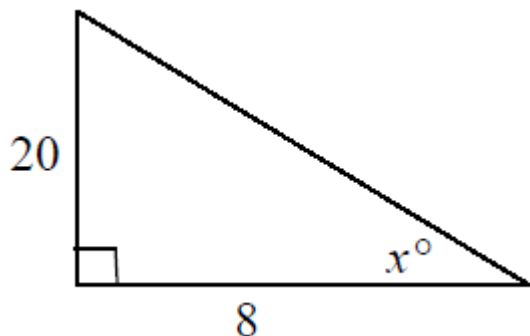


We are given the angle, the adjacent and the hypotenuse so that this is a cosine problem.

$$\cos x = \frac{3}{12} \quad \text{set-up trig ratio}$$

$$x = \cos^{-1}\left(\frac{3}{12}\right) \quad \text{use the inverse trig function to find angle}$$

$$x = 76^\circ \quad \text{put in calculator}$$



We are given the angle, the opposite and the adjacent so that this is a tangent problem.

$$\tan x = \frac{20}{8} \quad \text{set-up trig ratio}$$

$$x = \tan^{-1}\left(\frac{20}{8}\right) \quad \text{use the inverse trig function to find angle}$$

$$x = 68^\circ \quad \text{put in calculator}$$

Soh - Cah - Toa:

Intoduction to Trigonometry Notes

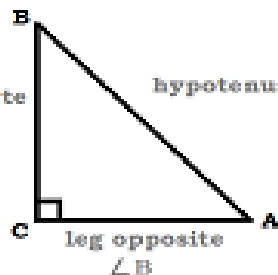
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The ratio of the measure of the leg opposite of an angle to the hypotenuse in a right triangle



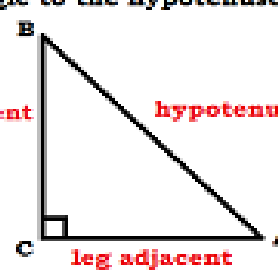
$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin A = \frac{BC}{BA}$$

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Cosine

The ratio of the measure of the leg adjacent to an angle to the hypotenuse in a right triangle



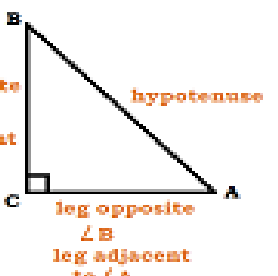
$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos A = \frac{CA}{BA}$$

$$\cos B = \frac{BC}{BA}$$

Tangent

The ratio of the measurement of the leg opposite an angle to the measurement of the leg adjacent in a right triangle



$$\tan = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan A = \frac{BC}{CA}$$

$$\tan B = \frac{CA}{BC}$$

SOH – CAH – TOA

S – Sine

O – Opposite

H – Hypotenuse

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$\tan \angle L$

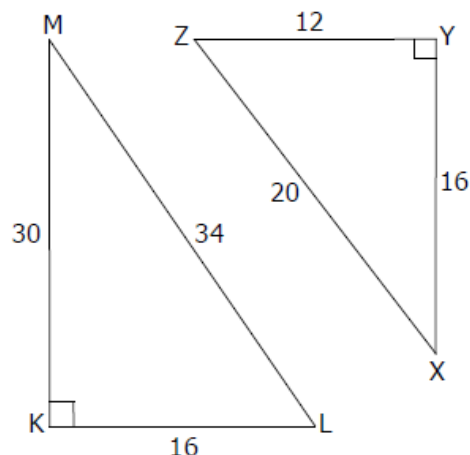
$\tan \angle X$

$\cos \angle L$

$\tan \angle Z$

$\cos \angle M$

$\sin \angle X$



Using the calculator:

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$$\tan 74^\circ$$

$$\sin 7^\circ$$

$$\cos 24^\circ$$

$$\tan 54^\circ$$

$$\sin 72^\circ$$

$$\cos 12^\circ$$

$$\cos 20^\circ$$

$$\tan 37^\circ$$

Examples: Find the measure of each angle to the nearest whole degree.

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$$\tan C = 0.4279$$

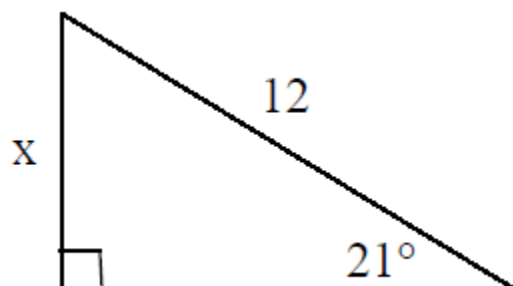
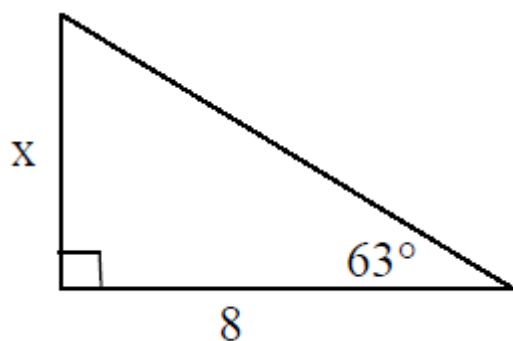
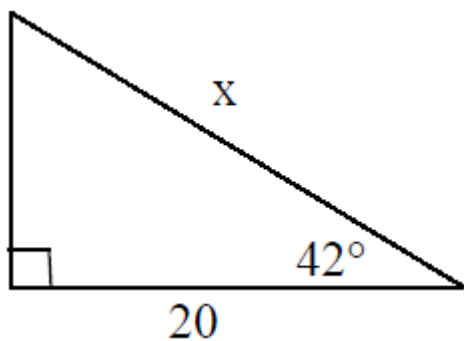
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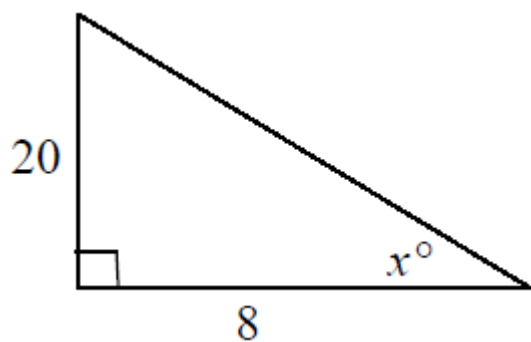
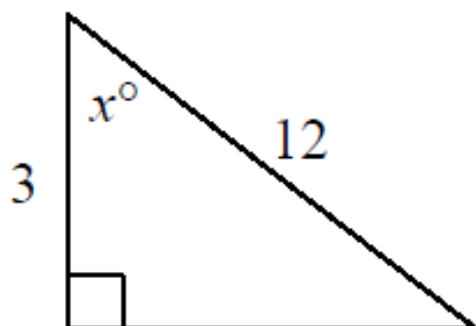
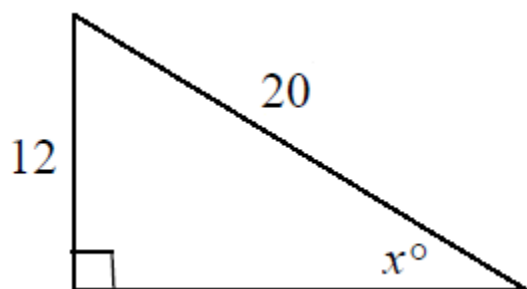
$$\text{trig ratio (angle)} = \frac{\text{side}}{\text{side}}$$

- They will be solved in different ways however.

Finding the Missing Side Problems:



Finding the Missing Angle Problems:



Name: _____ **ANSWER KEY** _____ Hour: _____ Date: _____

Soh - Cah - Toa:

Intoduction to Trigonometry Assignment

Part one: Find the ratio for each given Trig. Function.

$$\sin \angle A = \frac{8}{20} = \frac{2}{5}$$

$$\cos \angle W = \frac{30}{63} = \frac{10}{21}$$

$$\sin \angle P = \frac{30}{63} = \frac{10}{21}$$

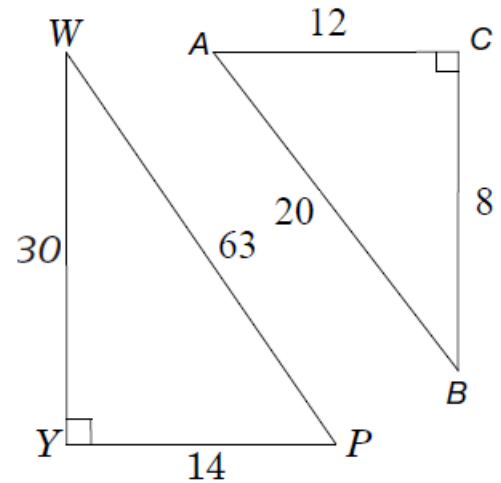
$$\cos \angle A = \frac{12}{20} = \frac{3}{5}$$

$$\tan \angle A = \frac{8}{12} = \frac{2}{3}$$

$$\cos \angle B = \frac{8}{20} = \frac{2}{5}$$

$$\sin \angle B = \frac{12}{20} = \frac{3}{5}$$

$$\tan \angle W = \frac{14}{30}$$



Part Two: Find the value of each ratio. Round your answer to 4 places past the decimal place.

$$\sin 51^\circ = 0.7771$$

$$\cos 47^\circ = 0.6820$$

$$\tan 9^\circ = 0.1584$$

$$\sin 26^\circ = 0.4384$$

$$\tan 89^\circ = 57.2899$$

$$\cos 1^\circ = 0.9998$$

Part Three: Find the measure of each angle to the nearest whole degree.

$$\sin A = 0.9876 = 81^\circ$$

$$\cos B = 0.8829 = 28^\circ$$

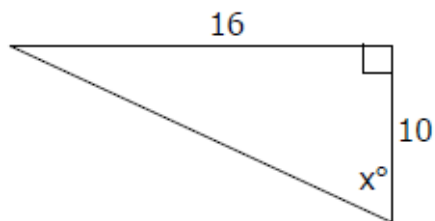
$$\tan S = 0.0349 = 2^\circ$$

$$\sin T = 0.0523 = 3^\circ$$

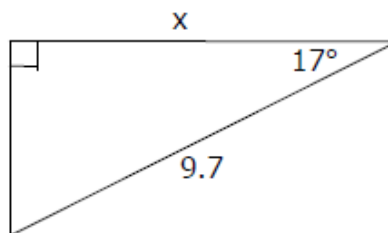
$$\cos X = 0.4848 = 61^\circ$$

$$\tan U = 28.6362 = 88^\circ$$

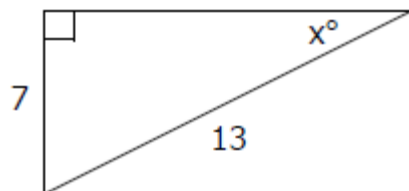
Part Four: Find the value of “x”. Round sides to the nearest tenth and angles the nearest whole degree.



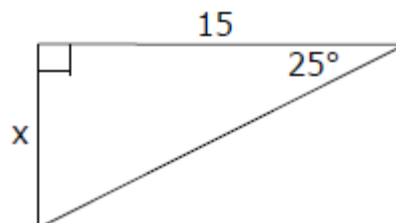
$$x = 58^\circ$$



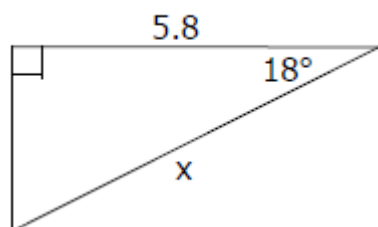
$$x = 9.28$$



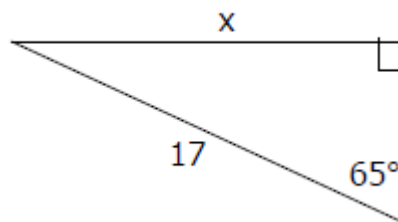
$$x = 33^\circ$$



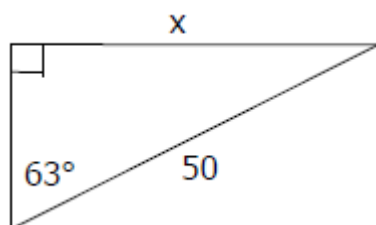
$$x = 6.99$$



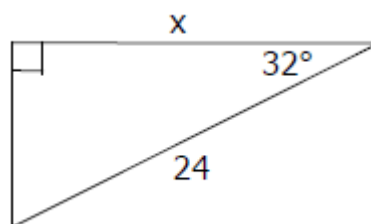
$$x = 58^\circ$$



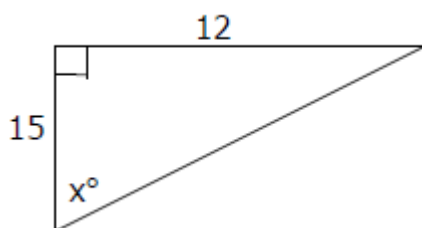
$$x = 15.41$$



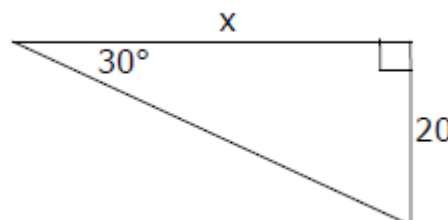
$$x = 44.55$$



$$x = 20.35$$



$$x = 39^\circ$$



$$x = 34.64$$

Name: _____ Hour: _____ Date: _____

Soh - Cah - Toa:

Introduction to Trigonometry Assignment

Part one: Find the ratio for each given Trig. Function.

$\sin \angle A$

$\cos \angle W$

$\sin \angle P$

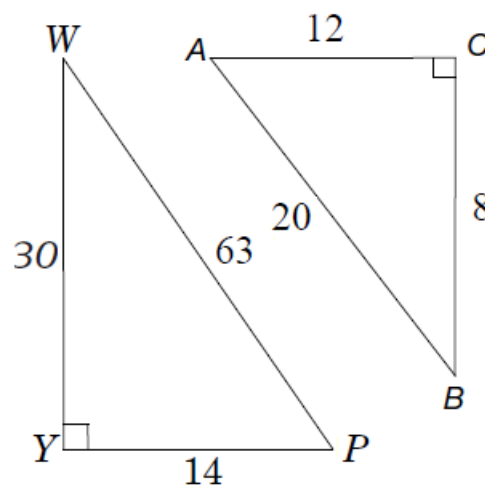
$\cos \angle A$

$\tan \angle A$

$\cos \angle B$

$\sin \angle B$

$\tan \angle W$



Part Two: Find the value of each ratio. Round your answer to 4 places past the decimal place.

$\sin 51^\circ$

$\cos 47^\circ$

$\tan 9^\circ$

$\sin 26^\circ$

$\tan 89^\circ$

$\cos 1^\circ$

Part Three: Find the measure of each angle to the nearest whole degree.

$\sin A = 0.9876$

$\cos B = 0.8829$

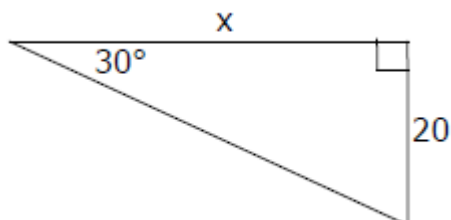
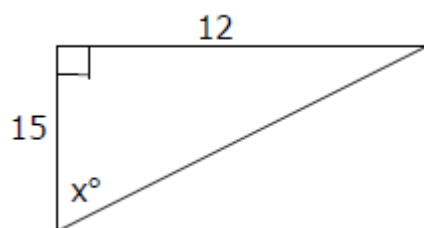
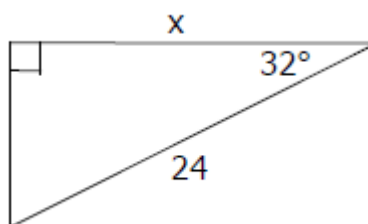
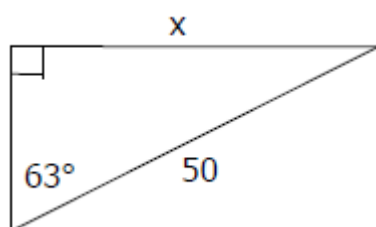
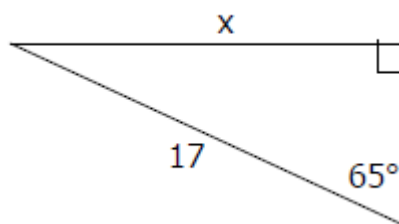
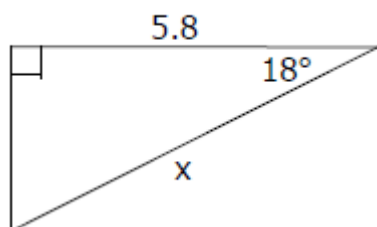
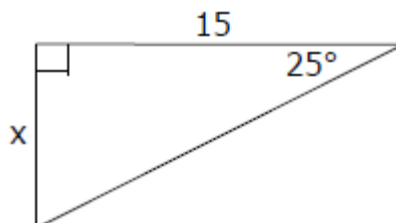
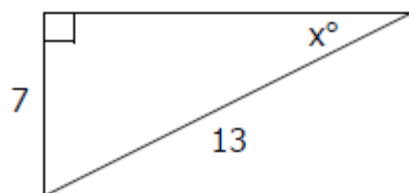
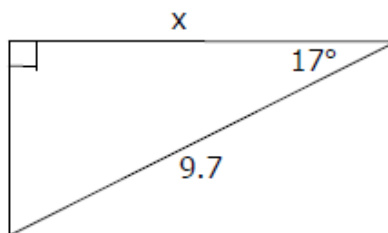
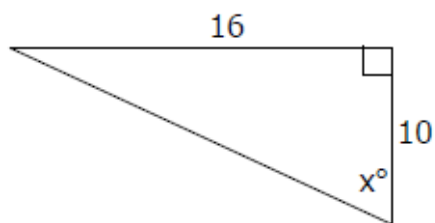
$\tan S = 0.0349$

$\sin T = 0.0523$

$\cos X = 0.4848$

$\tan U = 28.6362$

Part Four: Find the value of “x”. Round sides to the nearest tenth and angles the nearest whole degree.



Thank You For Your Purchase!

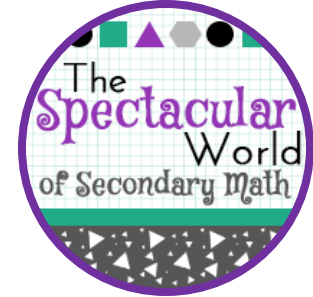
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